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Roll No

AU-8001 (CBGS)

B.E. VIII Semester Examination, June 2020

Choice Based Grading System (CBGS)

Vehicle Dynamics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.
iii) Draw neat sketches wherever necessary.

1. a) What do you mean by Vehicle control loop? 7
b) What is multi-body system approach? Where it is used? 7
2. a) Draw a neat sketch to show the various tire forces and moments. 7
b) What do you mean by ride properties of Tires? Enlist them. 7
3. Derive the equations for braking performance of a four wheeled vehicle moving on a plane road. 14
4. a) What is direction stability? How tires effect it? 7
b) What do you mean by transient response characteristics? Enlist a few. 7
5. a) Discuss the various vehicle ride models. 7
b) Explain the utility of frequency response function. 7
6. a) Enlist the various Stability and handling characteristics for a two wheeler. 7
b) Explain gyroscopic effect for a two wheeler on mathematical ground. Is it helpful? 7
7. a) Discuss the physics of tire on physical and mathematical grounds. 7
b) Discuss mathematically the effect of mass distribution on vehicle stability. 7
8. a) Write a short note on operating fuel economy. 7
b) Discuss mathematically how road surface profile acts as a random function. 7
